Octahedral cuts and chains of ideal right-angled 24-cells

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Abstract

We study a restricted class of finite-volume ideal right-angled hyperbolic four-polytopes: every facet must be either a regular ideal octahedron or the double of one across a triangular face. Inside each doubled facet there is a distinguished ideal triangle. We show that these triangles fit together into closed surfaces, and that every connected surface consists of exactly eight triangles. Equal cusp widths then put its triangles in one hyperplane and produce a regular ideal octahedral section. Cutting along all such sections decomposes the polytope into regular ideal 24-cells. An additional argument in the D_4 root model forces the cells to form a chain, with consecutive cuts opposite in each intermediate cell. The facet hypothesis is essential: no classification of arbitrary ideal right-angled four-polytopes or universal volume quantization is claimed. The proof is written in full, with the geometric and topological reductions separated from the finite model calculations. It is an internal mathematical argument; external review and a complete Lean formalization remain outstanding.

1 The precise problem

Let $\mathcal{O}$ be a regular ideal right-angled octahedron in $\mathbb{H}^3$. Reflect $\mathcal{O}$ across one triangular face and adjoin the reflected copy. We denote the resulting ideal right-angled three-polyhedron by $\mathcal{D}$. It has nine vertices, eight triangular faces and three quadrilateral faces. Let $\mathcal{C}$ be the regular ideal right-angled 24-cell in $\mathbb{H}^4$.

Throughout this paper, P is a finite-sided, convex, finite-volume four-polytope. Every vertex of P is ideal and every dihedral angle is $\pi/2$. We impose the additional condition

$$\text{every facet of } P \text{ is isometric to } \mathcal{O} \text{ or } \mathcal{D}. \tag{1}$$

The word "isometric" fixes the intrinsic geometry of the facets, not merely their face counts. One can instead assume the corresponding combinatorial types and invoke the usual three-dimensional right-angled rigidity, but no such classification is needed once (1) is part of the hypothesis.

Theorem 1.1. *Suppose that P satisfies (1). There are finitely many pairwise disjoint regular ideal octahedral sections which cut P into regular ideal right-angled 24-cells. If P has N facets of type $\mathcal{D}$, then $N = 8k$ for some integer $k \geq 0$, and there are $k+1$ cells in the decomposition. Their adjacency graph is a path. In every cell incident to two cuts, the two cutting facets are opposite. Consequently, within the class (1), the isometry type of an existing polytope is determined by its number of cells, and*

$$\operatorname{Vol}(P) = (k+1)\frac{4\pi^2}{3}.$$

The last assertion classifies existing polytopes in this restricted class; we do not use it to assert existence for every prescribed path length. In particular, Theorem 1.1 is not a proof that every ideal

right-angled four-polytope is a chain of 24-cells. Its role is to identify exactly how far the octahedral cutting argument extends without allowing more complicated facets.

The proof has two distinct stages. We first build a cutting section from intrinsic facet geometry. We then control the arrangement of the sections inside the resulting 24-cells. Neither stage assumes that the polytope was constructed by gluing cells.

2 What a doubled facet contributes at a cusp

The three vertices of the gluing face of $\mathcal{D}$ have a simple intrinsic description: their horospherical rectangles are nonsquare. At each of these vertices the rectangle is obtained by adjoining two unit squares along one side, and therefore has aspect ratio two. The remaining six rectangles are square. The three quadrilateral faces of $\mathcal{D}$ meet pairwise at its three nonsquare cusps and have no common edges. At a nonsquare cusp, the two quadrilateral sides are the two long sides of its 2 by 1 rectangle.

The convex hull of the three nonsquare cusps is an ideal triangle T_F whose relative interior lies in the relative interior of a doubled facet F. This is its original gluing triangle. Each side of T_F is the interior diagonal of a quadrilateral face of F. At a nonsquare cusp, T_F meets the horospherical rectangle along the transverse segment halfway along its long direction. Figure 1 records these facts in the Euclidean link, where their metric meaning is literal.

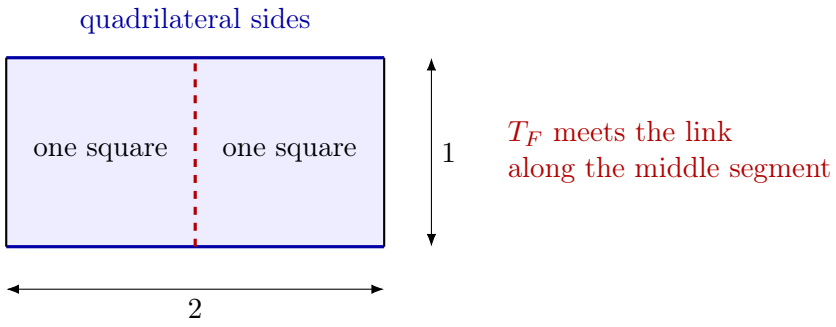


Figure 1: The cusp rectangle of a doubled octahedron at a gluing vertex. The blue sides belong to its quadrilateral faces. The red segment is the link of the internal triangle. This is an exact Euclidean link diagram, not a perspective drawing of $\mathbb{H}^4$.

At a square cusp of $\mathcal{D}$, exactly one incident polygonal face is a quadrilateral. At a cusp of $\mathcal{O}$, none is. These two small incidence facts will exclude one possible ambient box shape.

Lemma 2.1. *Every noncubic cusp box of P has side proportions $2:1:1$. Its four nonsquare faces belong to doubled facets and form a belt; its two square end faces belong to ordinary octahedral facets. Every adjacency within the belt is a quadrilateral ridge of P.*

Proof. Each face rectangle of a cusp box is intrinsic to a facet of P. Its aspect ratio is therefore one or two. If the three side lengths were distinct, the ratio of the largest to the smallest would exceed two, since the two consecutive unequal ratios must both be two. Thus a noncubic box has proportions $2:1:1$ or $2:2:1$.

Consider a $2:2:1$ box. The four nonsquare faces must be doubled facets. In each, the two long sides are quadrilateral sides. Each square end face consequently has four incident quadrilateral sides. An octahedral cusp has none, and a square cusp of a doubled facet has exactly one. Neither permitted facet type can supply an end face. The $2:2:1$ box is impossible.

In a $2:1:1$ box, the four long rectangles are the belt. Their mutual sides are the long sides and hence are quadrilateral ridges. The square ends meet the belt only along its short, triangular sides. All four incidences at an end are triangular, so the end cannot be a square cusp of a doubled facet. It is octahedral. □

3 The internal triangles form closed surfaces

Join two doubled facets when they share a quadrilateral ridge. The distinguished triangles in these facets have the same internal diagonal in that ridge, so we glue them along this diagonal. The gluing is geometrically defined; there is no choice of an abstract edge identification.

Lemma 3.1. *After adjoining their ideal vertices, the glued triangles form a finite closed two-manifold. Its connected components are exactly the components of the quadrilateral adjacency graph of doubled facets. Every edge lies in two triangles and every vertex lies in four.*

Proof. The interior of a distinguished triangle lies in the relative interior of its facet. Interiors of different facets are disjoint. An edge of the triangle is in the relative interior of an actual quadrilateral ridge, which lies in precisely two facets. Thus each edge has two incident triangles and has a disk neighborhood at each of its finite points.

It remains to check the ideal vertices. By Lemma 2.1, the triangles incident to one such point come from the four doubled facets in a single $2:1:1$ belt. Their four middle segments join end to end around the box, giving the simple square shown in Figure 2. They do not cross and do not leave an open end. This square is the link of the point in the glued surface. Coning the link gives a disk after the ideal point is adjoined. There are no other triangle vertices.

All four doubled facets at such a vertex are already connected through their quadrilateral belt. Therefore identifying ideal vertices does not join two different components of the quadrilateral adjacency graph. The components of the graph and of the surface coincide. □

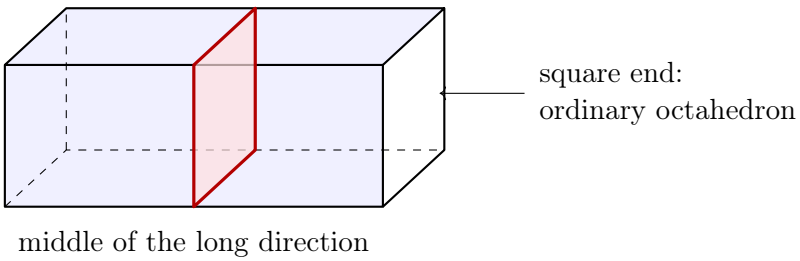


Figure 2: Four internal triangles meet an ambient $2:1:1$ cusp box in its transverse middle square. Its boundary is the four-segment link used to prove that the compactified triangle complex is a two-manifold. The filled square indicates the transverse section; only its boundary lies on the box surface.

Lemma 3.2. *A connected component containing n triangles contains exactly eight triangles, twelve edges and six vertices.*

Proof. Write e and v for its edge and vertex counts. The incidence counts of Lemma 3.1 give

$$e = \frac{3n}{2}, \qquad v = \frac{3n}{4}, \qquad \chi = v - e + n = \frac{n}{4}.$$

We shall use only the elementary inequality $\chi \leq 2$ for a connected closed cellulated surface; we do not assume the surface is a sphere. Here is a direct proof in the present setting. Choose a spanning tree A of its one-skeleton. Delete from the dual graph the edges dual to A. The remaining dual graph is connected. Otherwise, the boundary between one nonempty proper set of triangular faces and its complement would be a nonempty subgraph of A. At each vertex its degree is even: a walk around the circular link enters and leaves the chosen set equally often. This gives a nonempty Eulerian subgraph of a tree, which is impossible. We may therefore choose a dual spanning tree disjoint from A. The two trees use $v-1$ and $n-1$ distinct edges, so $e \geq v+n-2$ and $\chi \leq 2$. It follows that $n \leq 8$, and the integer $\chi = n/4$ also shows that n is divisible by four.

We next obtain a lower bound within the same component. Choose one doubled facet F. Its three quadrilateral faces have three distinct doubled neighbors. Its three nonsquare cusps also have three opposite doubled facets. The latter are distinct from one another and from every ridge neighbor: their walls are asymptotic to the wall of F at exactly one specified ideal point, whereas a ridge neighbor intersects its wall along a plane. These opposite facets belong to the same component, because opposite belt members are connected by the four-cycle of quadrilateral adjacencies. Together with F we have found seven distinct facets in the component. Thus $7 \leq n \leq 8$ and $4 \mid n$, forcing $n = 8$. The values of e and v follow. □

4 From six cusps to eight wall normals

Fix one component. Make a graph J whose six vertices are its cusps and whose edges are the twelve internal diagonals. Every vertex has degree four.

Lemma 4.1. *The graph J is the octahedral graph. Its eight triangles are the eight distinguished facet triangles. The doubled facets can therefore be indexed by $s \in \{\pm 1\}^3$ so that quadrilateral adjacency changes exactly one sign.*

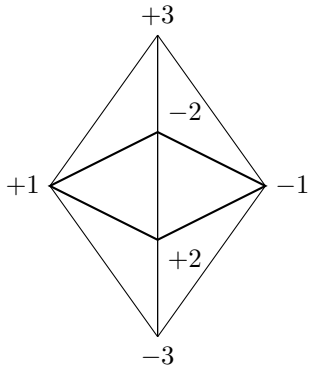
Proof. First, J has no parallel edges. If two different quadrilateral ridges joined the same two cusps, their incident facets would include at least three of the four belt members at either cusp. Among three belt members there are two opposite ones. Their walls are asymptotic at that cusp and cannot contain the second cusp. This is a contradiction. Two distinct ridges cannot have the same pair of incident facets, since the intersection of two convex facets is one face. Thus J is simple.

A simple four-regular graph on six vertices has complement a perfect matching, so it is the octahedral graph. Each distinguished triangle gives a three-cycle in J. Two facets cannot give the same three-cycle: its geometric triangle would then have an interior point in the relative interiors of two different facets. The octahedral graph has exactly eight three-cycles, obtained by choosing one vertex from each of its three opposite pairs. All eight occur. Label these opposite pairs by $1, 2, 3$, with a sign on each member. A triangle, and hence a doubled facet, is indexed by its three choices of sign. Two triangles share an edge exactly when one sign changes. □

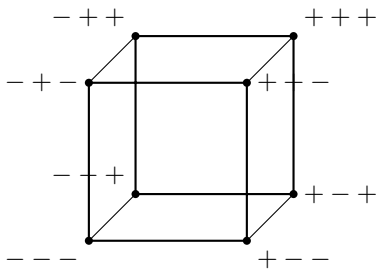
5 Equal widths put the triangles in one hyperplane

We work in $\mathbb{R}^{4,1}$ with form $\langle x, x\rangle = -x_0^2 + x_1^2 + \cdots + x_4^2$. A wall has an outward unit spacelike normal n, and its halfspace is $\langle x, n\rangle \leq 0$. Let n_s be the normals of the eight doubled facets just indexed.

Lemma 5.1. *On each square of the cube in Figure 3, the sums of the two opposite pairs of normals agree.*



cusps and diagonals



doubled facets and quadrilaterals

Figure 3: The two incidence graphs. The six-cusp graph is octahedral; its eight triangles index the doubled facets. The dual cube records quadrilateral adjacency. The signs specify which cusp from each opposite pair belongs to a distinguished triangle. These are incidence drawings, not metric projections of the polytope.

Proof. The square is a cusp belt. Its opposite wall pairs bound the two short directions of the $2:1:1$ box, so their Euclidean separations on a common horosphere agree. The use of unit normals matters. With cusp null vector $\xi = (1,0,0,0,1)$, the vertical wall $x_i = a$ in upper-half-space has unit normal (a, e_i, a). The opposite wall $x_i = b$, with the other outward orientation, has normal $(-b, -e_i, -b)$. Their sum is $(a-b)\xi$. Thus its coefficient is the separation, with the same normalization for both pairs. Equal widths give equal sums. □

Lemma 5.2. *There are mutually orthogonal vectors a, b_1, b_2, b_3 such that*

$$n_s = a + s_1 b_1 + s_2 b_2 + s_3 b_3, \qquad a^2 = -\tfrac{1}{2}, \quad b_i^2 = \tfrac{1}{2}.$$

Proof. The equations of Lemma 5.1 say that every mixed second finite difference on the cube vanishes. Expanding $s \mapsto n_s$ in sign characters leaves only its constant and three linear terms, giving the displayed affine expression. Expanding the eight equations $n_s^2 = 1$ and comparing sign characters gives

$$\langle a, b_i \rangle = 0, \quad \langle b_i, b_j \rangle = 0 \ (i \neq j), \quad a^2 + b_1^2 + b_2^2 + b_3^2 = 1.$$

Adjacent facets meet orthogonally. If s' differs from s only in coordinate i, then $0 = \langle n_s, n_{s'} \rangle = 1 - 2b_i^2$. Thus every $b_i^2 = 1/2$, and the remaining identity gives $a^2 = -1/2$. □

The span U of these four vectors is Lorentzian and has dimension four. Its orthogonal complement is a spacelike line. The intersection of U with $\mathbb{H}^4$ is a genuine hyperbolic hyperplane H, orthogonal to all eight doubled-facet walls. In orthonormal coordinates on U, the wall normals are $(1, s_1, s_2, s_3)/\sqrt{2}$, with the future sign determined by their positive cusp-null sums. Their halfspaces cut out

$$S = \{(t, x) \in \mathbb{H}^3 : |x_1| + |x_2| + |x_3| \leq t\}, \tag{2}$$

the regular ideal octahedron. Its six null rays are $a \pm b_i$. They are exactly the actual cusps of the component, because the opposite belt-normal sums are $2(a \pm b_i)$.

Convexity now supplies the step that an abstract graph cannot: since these six ideal points belong to the closure of P, their hyperbolic convex hull S is contained in P. Conversely the eight actual facet inequalities imply $P \cap H \subset S$. Therefore

$$P \cap H = S.$$

Each face of S is its facet's distinguished internal triangle. Each edge of S is the internal diagonal of an actual quadrilateral ridge, as illustrated in Figure 4.

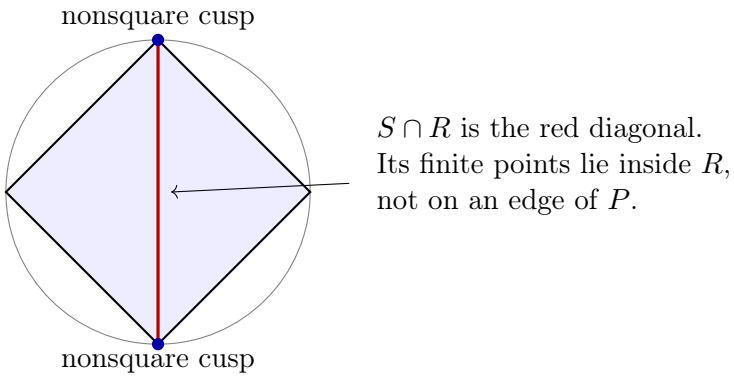


Figure 4: A quadrilateral ridge R in the Klein disk. Its two nonsquare cusps are opposite vertices. Their diagonal is an edge of the cutting octahedron. This distinction is what prevents an unrelated facet from touching the section at a finite edge point.

6 Why the hyperplane is a legitimate cut

Lemma 6.1. *No facet outside the chosen eight meets S at a finite point. Cutting along H gives two finite-volume entirely ideal right-angled four-polytopes. The eight doubled facets become ordinary octahedra; every other facet is uncut.*

Proof. Let F be any outside facet. Its supporting functional is nonpositive on S. If it vanishes at a relative interior point of a face of S, it vanishes on that face. A finite point of S lies in its interior, in a triangular face interior, or in an edge interior. A triangular face interior belongs to the relative interior of one of the chosen doubled facets, so it cannot belong to F. An edge interior lies in the relative interior of a ridge whose only two incident facets are the chosen doubled facets. It cannot belong to F either. A supporting functional vanishing at an interior point would contain all of S, contradicting the face-interior case. Thus F has no finite intersection with S.

If F crossed H, a finite segment in F would meet $P \cap H = S$ at a finite point. Consequently no outside facet crosses. The chosen eight facets are cut along their genuine gluing triangles into octahedra. The center of S is strict on all eight wall inequalities and, by what was just proved, on every other wall inequality. It is an interior point of P; both halves have nonempty interior.

In the Klein model, a vertex of a half lying on H must be a vertex of S. Otherwise it lies in a nontrivial segment of S and is not extreme. Away from H, a vertex is an old vertex of P. The only vertices of S are its six old ideal cusps, so no finite vertex is introduced. The new facet S meets only the eight cut walls, all orthogonal to H. Every new dihedral angle is right; all other angles are inherited. Finite volume follows by containment in P. □

The facet spectrum (1) is preserved by the cut. There are exactly eight fewer doubled facets, counted over both pieces. Repeating the construction therefore terminates.

There is also a useful geometric disjointness statement. Every outside doubled facet lies on one side of the first cut. Two quadrilateral-adjacent outside facets cannot lie on opposite sides, since their shared ridge would give a finite intersection with S. Thus a whole other component lies on one side. Its section, being the convex hull of its six cusps, lies on that side as well. A noncubic

cusp belongs to only one belt component and has octahedral ends; two sections cannot share an ideal vertex. The outside-facet argument excludes finite boundary contact. Hence the sections are pairwise disjoint, including their ideal endpoints.

7 Recognizing a piece with only octahedral facets

We give a direct recognition argument rather than using a four-dimensional classification as a final black box.

Lemma 7.1. *An entirely ideal right-angled four-polytope whose facets are all isometric to $\mathcal{O}$ is isometric to $\mathcal{C}$.*

Proof. Place one facet F on a facet of a model regular 24-cell. Its eight triangular faces agree with those of the model facet. For any one of these triangles, the adjacent supporting wall is uniquely determined by the triangle and orthogonality to F's wall: its outward normal is the intrinsic outward unit normal of that triangle within F's wall.

In this adjacent wall, its octahedral facet is determined by the shared ideal triangle and the side on which it lies. To justify this last assertion, take any isometry between two regular ideal octahedra with that triangular face. Every isometry of the face extends to a symmetry of the octahedron, so the isometry can be adjusted to fix the face vertices pointwise. An isometry of $\mathbb{H}^3$ fixing three distinct ideal points is the identity or reflection in their plane; the specified side selects the identity. The neighboring facet is therefore the whole model octahedron, not only a piece on the same wall.

Repeat across its faces. The facet adjacency graph of the model 24-cell is connected. The propagation matches all 24 model facets, with their full vertex sets, to actual facets of the piece. Their halfspaces put the piece inside the model cell. Conversely the closure of the piece contains every ideal model vertex, and convexity contains their hull, the model cell. The two polytopes agree. □

Each final piece is consequently a 24-cell. If the original polytope has N doubled facets, Lemma 3.2 gives $N = 8k$. Each of the k cuts increases the number of pieces by one, so there are $k+1$ cells. Their adjacency graph is a tree: starting with one piece, each separating cut replaces one vertex of the current adjacency tree by two vertices joined by one edge.

8 Why the tree is a chain

The restriction on facet types has one more consequence. In the Lorentz model a regular 24-cell has outward unit normals

$$(1, r), \qquad r \in \{(\pm 1, \pm 1, 0, 0) \text{ and all coordinate permutations}\}.$$

The spatial vectors are the roots of D_4. Distinct roots have dot products $-2, -1, 0, 1$, with -2 only for opposite roots.

Lemma 8.1. *If a cell in the decomposition meets two cutting sections, these are opposite facets of that cell.*

Proof. The sections have disjoint closures. If their roots are r_1, r_2, then $r_1 \cdot r_2$ is negative: dot product one gives an actual common triangular ridge, and dot product zero gives a common ideal vertex. Thus the dot product is -1 or -2.

Suppose it is -1. The vector $r_1 + r_2$ is another D_4 root, and its cell facet Q is orthogonal to both cut walls, since

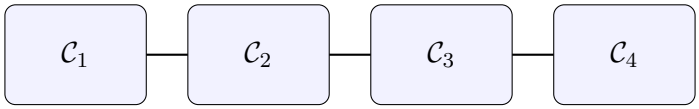
$$-1 + (r_1 + r_2) \cdot r_i = 0 \qquad (i = 1, 2).$$

It cannot itself be a cutting facet, since it meets the two sections along triangles. It is part of an original facet of P. Across each of the two cuts there is a neighboring regular 24-cell. Its corresponding facet is the reflection of Q and lies in the same original supporting wall, because that wall is orthogonal to the cut. This neighboring facet cannot be another internal cut: it meets the first cut along a triangle, whereas all cuts have disjoint closures. Thus the original facet contains three regular ideal octahedra.

Their relative interiors are disjoint. If two had an open overlap, a point of that overlap avoiding every other cell wall would have an inward normal collar contained in both cells. Their four-dimensional interiors would overlap, which is impossible for pieces obtained by cuts. The original facet has intrinsic volume at least three times $\mathrm{Vol}(\mathcal{O})$; it can be neither $\mathcal{O}$ nor $\mathcal{D}$. This contradicts (1). Hence the dot product is -2, and the two facets are opposite. □

No cell can have three pairwise opposite facets. The adjacency tree has maximum degree two and is a path. Once an end cell and its first gluing facet are placed, every next cell is its reflection across the gluing facet, and the next gluing facet in an intermediate cell is uniquely the opposite one.

For completeness, there is no independent gluing twist. The regular 24-cell is flag-transitive. Given an isometry of an octahedral facet, send one facet flag to its image by a cell symmetry. This symmetry stabilizes the facet and has the required restriction, because an octahedral isometry fixing a flag is the identity. Thus every facet symmetry extends to the cell. Two cells on opposite sides of the same facet are therefore uniquely placed up to the ambient reflection. This proves the isometry assertion in Theorem 1.1.



two cuts in an intermediate cell
must use opposite octahedral facets

Figure 5: The resulting adjacency graph is a path. The boxes denote whole 24-cells and are not metric four-dimensional drawings. The opposition condition comes from the common-neighbor argument in Lemma 8.1, rather than from the drawing.

9 Volume, face counts, and the limit of the result

The regular ideal 24-cell has volume $4\pi^2/3$. Volume additivity proves the formula in the main theorem. There is a second way to see the same integer. Write v, e, r, f for the face counts of P and put

$$W = \sum_R (k(R) - 3),$$

where $k(R)$ is the number of vertices of a ridge. Cusp-box incidence and Euler's equation give

$$e = 4v, \quad r = 3v + f, \quad W = 3(v - f).$$

An octahedral facet has defect zero and a doubled facet defect three, where the defect is the sum of $k(R) - 3$ over its faces. Every ridge is counted twice, so $2W = 3N = 24k$. The volume formula [1, Lemma 2] therefore gives

$$\mathrm{Vol}(P) = \frac{4\pi^2}{3} + \frac{\pi^2}{9}W = (k+1)\frac{4\pi^2}{3}.$$

If there are $m = k + 1$ cells, the complete face counts are

$$(v, e, r, f) = (18m + 6,\ 72m + 24,\ 68m + 28,\ 14m + 10).$$

Indeed each gluing identifies six ideal vertices and removes two facets while merging eight neighboring pairs. There are no extra vertex identifications between nonadjacent cells: a common cusp would lie on every separating cut along their path, whereas different sections have no common ideal vertex. The edge and ridge formulas then follow from the incidence identities above. There are $8(m - 1)$ doubled facets, $6m + 18$ ordinary facets and $12(m - 1)$ quadrilateral ridges. All other ridges are triangles. For $m = 2$ this recovers $(42, 168, 164, 38)$.

These formulas do not establish the same quantization for arbitrary ideal right-angled four-polytopes. Without (1), a facet may have more complicated cusp rectangles, the internal triangles used here may not exist, and the local links need not be the four-segment belts of Lemma 3.1. A successful extension must replace those geometric facts, not merely repeat the Euler count.

A Proof inputs and verification status

The geometric inputs are the explicit regular octahedron, its double, the regular 24-cell, the rectangular link of an ideal right-angled cusp, and the usual Lorentz description of supporting walls. The surface Euler bound, cusp-component injection, six-cusp graph, normal equations, actual cutting argument, all-octahedral recognition and opposition argument are proved in this manuscript. The volume formula is cited for the alternative computation; the decomposition and volume-additivity argument do not depend on a published volume-minimum classification.

The written decomposition proof has a separate internal mathematical review. The stronger opposition argument is subject to its own review record. Existing exact local certificates check the octahedron-double cusp data and cube-normal algebra; they do not certify the universal surface construction or convexity arguments. No end-to-end Lean verification or external human review is claimed. Build records and the precise review links belong to the reproduction document, rather than serving as mathematical premises here.

References

[1] A. Kolpakov, *On the optimality of the ideal right-angled 24-cell*, Algebraic & Geometric Topology **12** (2012), 1941–1960. doi:10.2140/agt.2012.12.1941.